

Experiment No. 5

Inorganic Qualitative Analysis (Salt no.1)

Aim: Analyse one basic (cation) and one acidic (anion) radical from given inorganic salt.

Apparatus: Test tubes, test tube stand, spatula, glass rod, watch glass, beaker, burner, etc.

A. Preliminary tests: (General scheme)

Test	Observation	Inference
Nature	Crystalline (soluble in water).	Cl^- , Br^- , I^- , NO_3^- , SO_4^{2-} etc. may be present.
Color of salt	White or colourless	Pb^{2+} , Al^{3+} , Mn^{2+} , Zn^{2+} , Ba^{2+} , Ca^{2+} , Sr^{2+} , Mg^{2+} , NH_4^+ may be present.
Solubility	Soluble in water	Crystalline salt present.

B. Dry tests for basic radicals (Cations)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of salt in a clean, dry test tube and heat it strongly. Observe the color of the residue and gas evolved.	White sublimate	NH_4^+ may be present.
2. Test for NH_4^+ Salt + NaOH, heat gently, hold moist turmeric paper near the mouth of test tube.	Smell of NH_3 gas turning moist turmeric paper reddish brown	NH_4^+ may be present.
3. Charcoal cavity test: Take the salt and solid Na_2CO_3 in the ratio 1:2 on watch glass, add few drops of water and prepare a paste, place this in charcoal cavity and heat it with a blow pipe in a reducing flame.	—	—
4. Cobalt Nitrate Test: If white infusible residue is obtained in charcoal cavity test, add two drops of cobalt nitrate solution on the white residue and heat in the blue flame.	—	—

5. Flame test: Clean a platinum wire with conc. HCl. Moisten a small quantity of the substance with concentrated HCl on watch glass. Dip the cleaned platinum wire in it and heat it in oxidizing flame (non - luminous) and observe the color of the flame		
6. Borax bead test: [For colored salt only] Make a small loop at the end of a clean platinum wire heat it strongly and dip it in borax powder. Heat it again till a glossy bead is obtained. Now touch the hot bead with a small crystal of the salt and heat it again in oxidizing (blue) flame. Observe the color of the bead.		

C. Dry tests for acidic radicals (Anions)

Test	Observation	Inference
i. Action of dil HCl: Salt + dil HCl.	Colourless gas with brisk effervescence.	CO_3^{2-} absent.
ii. Action of conc. H_2SO_4: Salt + conc. H_2SO_4 + heat	Colourless gas giving white dense fumes glass rod dipped in NH_4OH .	Cl may be Present.
iii. Action of MnO_2 and conc. H_2SO_4: Salt + a pinch of MnO_2 + conc. H_2SO_4 and heat	Colourless gas, yellow gas turning moist due to blue litmus paper and then bleaches.	Cl may be Present.
iv. Cu-foil test: (If brown gas is obtained in the above test then only perform this test) Salt + Cu-foil + conc. H_2SO_4 and heat strongly		

Note: (If the anion is not detected in above dry test, then salt may contain SO_4^{2-} or PO_4^{3-})

D. Preparation of solution (Original Solution): Dissolve about 0.5 gm of salt in 20 mL of water in a 50 mL beaker. This is called original solution (O.S.)

E. Wet test for basic radical (cations)

Detection of groups

Test	Observations	Inference
1. O.S. + dil. HCl	No ppt	Group I absent
2. O.S. + dil. HCl + H ₂ S gas / H ₂ S water solution	No ppt	Group II absent
3. O.S. + NH ₄ Cl + NH ₄ OH till alkaline	No ppt	Group IIIA absent
4. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + H ₂ S gas / H ₂ S water solution	No ppt	Group IIIB absent
5. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + saturated solution of (NH ₄) ₂ CO ₃	No ppt	Group IV
6. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + Na ₂ HPO ₄	No ppt	Group V absent
7. If all the above groups are absent	Hence	Group VI (NH ₄ ⁺) may be present.

Analysis of Group: After detection of a particular group do analysis of that group.

Analysis of Group

Test	Observations	Inference
O.S + NaOH Solution Heat.	Smell of NH ₃	NH ₄ ⁺ Present
C.T. for..NH₄⁺..(Ammonium) 1. O.S + 2 drops of Nessler's reagent.	Brown Precipitate	NH ₄ ⁺ confirmed
2. O.S + Picric acid	Yellow ppt	NH ₄ ⁺ confirmed

F. Wet test for acidic radicals:

Detection of acidic radical from water solution:

1. Wet tests for CO_3^{2-} , SO_3^{2-} , Cl^- , Br^- , I^-

1. O.S. + AgNO_3	White Ppt insoluble in dil. HNO_3 .	Cl^- , Br^- , I^- are present.
---------------------------	--	---

Distinction between halides (Cl^- , Br^- , I^-):

Note: (Perform this test only if ppt is obtained for AgNO_3 which is insoluble in dil. HNO_3 .)

Test	Observation	Inference
O.S. + Chloroform + Cl_2 water in excess (shake well)	chloroform layer is present.	Cl^- present.

Note: (If one of the halide radical is present then further analysis is not necessary. Perform confirmatory test for that detected radical.)

2. Wet test for SO_4^{2-} and PO_4^{3-}

(This test is required to perform only, if ppt is NOT obtained in AgNO_3 test)

Test	Observation	Inference
O.S. + $\text{Ba}(\text{NO}_3)_2$	_____	_____

3. WET TEST FOR NO_2^-

Test	Observation	Inference
O.S. + dil. H_2SO_4 + FeSO_4 solution	_____	_____

4. WET TEST FOR NO_3^-

Test	Observation	Inference
O.S. + 3-4 drops of conc. H_2SO_4 + 2-3 drops of diphenyl amine	_____	_____

5. WET TEST FOR CH_3COO^-

Test	Observation	Inference
O.S. + FeCl_3 solution	_____	_____

6. WET TEST FOR $\text{C}_2\text{O}_4^{2-}$

Test	Observation	Inference
O.S. + dil. Acetic acid + CaCl_2	White ppt is obtained	2CO_3^{2-} Present.

C.T. for

Test	Observation	Inference
1. 0.5 + MnO_2 + 2,3 drops of conc. H_2SO_4	Greenish yellow gas evolves that turns moist blue litmus red and then bleaches it.	Cl^- confirmed.
2. 0.5 + lead acetate.	White ppt	Cl^- Confirmed.

Result:- Given inorganic salt no. 1 contains following

- Basic radical NH_4^+
- Acidic radical..... Cl^-
- Molecular formula of the salt..... NH_4Cl
- Hence the name of the salt..... Ammonium chloride

Remark and sign of teacher: